K L UNIVERSITY

Department of Electronics and Communication Engineering

CAPSTONE PROJECT REVIEW PROGRESS RECORD

Review Stage: Review–3	**Date of Review: 06 / 11 / 2026**
Project Title:	Sensitivity-Guided Layer-Wise Approximate Multiplier Mapping for FPGA-Based CNN Accelerators
Cohort / Batch:	2024 – 2028 (Batch B4)
Project Guide:	Dr. Arun Nimmala, Associate Professor, Dept. of ECE

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Brief Description of the Project

Briefly explain the premise of the project, the problem being addressed and the proposed approach. (less than 200 words)

Deep Convolutional Neural Network (CNN) inference on resource-constrained edge hardware is dominated by Multiply-Accumulate (MAC) units, where 8×8 multipliers consume substantial silicon area and power (68 LUTs on Xilinx Zynq-7020). Approximate computing offers significant energy and area reduction by relaxing arithmetic precision. However, uniform approximation across an entire network causes severe accuracy degradation or yields negligible hardware savings. Existing layer-wise mapping techniques require multi-day genetic search (e.g., ALWANN requires 7.5 CPU-days) or costly GPU retraining loops. This project develops a lightweight, single-pass sensitivity scoring metric computed in just 0.2 seconds without retraining or iterative search. By coupling static weight variance with dynamic first-order loss gradients extracted across 256 calibration images, our framework accurately quantifies layer-wise error resilience. Error-critical early layers (conv1) are allocated exact multipliers, intermediate layers receive compensated designs, and noise-resilient deep layers (conv4, accounting for 48.5% of total MACs) are assigned aggressive truncated multipliers. The heterogeneous datapath is realized as a synthesizable AXI4-Stream accelerator IP core for physical deployment on the Xilinx Zynq-7020 (PYNQ-Z2) FPGA SoC, maximizing silicon efficiency while preserving baseline classification accuracy.

Progress Achieved up to This Review

Briefly mention the work completed, implementation/simulation/experimental progress, results obtained and other major milestones achieved up to this review.

1. **Complete Synthesizable Accelerator RTL Microarchitecture:** Engineered a fully parameterizable CNN accelerator IP in synthesizable Verilog (`cnn_accelerator_top.v`). Implemented a 2-bit dynamic mode controller (`layer_controller.v`), a signed MAC unit with sign-magnitude math (`mac_unit.v`), a 3×3×Cin Processing Element with finite-state control (`conv_pe.v`), and the unified multi-mode multiplier (`axm_8x8.v`).
2. **Exhaustive 262,144-Vector RTL Verification (100% Pass):** Developed an automated verification suite (`verify_rtl_logic.py`) exhaustively simulating all 4 multiplier modes across all 65,536 input combinations. Verified 262,144/262,144 vectors with zero functional mismatches against golden mathematical models.
3. **Vivado SoC Integration & Bitstream Automation:** Authored complete Vivado automation scripts (`vivado_synth.tcl` and `vivado_block_design.tcl`). Partitioned the system into Processing System (dual ARM Cortex-A9) and Programmable Logic (accelerator IP clocked at 100 MHz), interfaced via AXI DMA using high-performance HP0 ports for streaming 17-bit packed beats ({w_sign, w_mag, act_in}).
4. **PYNQ-Z2 On-Board Deployment & Jupyter Driver:** Developed the physical hardware driver (`pynq_inference.py`) and an interactive 7-cell Jupyter notebook (`pynq_cnn_demo.ipynb`). Pre-extracted INT8 quantized weights (`weights_int8.npy`) for direct DDR3 memory DMA transfers and real-time on-board latency/FPS measurement.
5. **Publication-Ready Deliverables & GitHub Repository:** Generated all 7 publication-grade figures (`fig1`–`fig7`) for the IEEE manuscript, authored a complete 20-slide technical presentation with embedded figures and full speaker notes, and established a fully documented, open-source GitHub repository with MIT License.

Signature of Project Guide: _____ **Date: 06 / 11 / 2026**